Phase Transitions in Backdoor Learning: Minimum Data Poisoning Thresholds for LLM Backdoors

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Abstract

Backdoor attacks pose a critical threat to the safety of large language models (LLMs), yet the minimum data poisoning required for successful backdoor implantation, the phase-change nature of this minimum data poisoning, and the degree of flexibility in the backdoor trigger have not been thoroughly studied. We present the first systematic dose-response study of backdoor activation in LLMs, finetuning 175 Llama 3.1 8B Instruct models using QLoRA across 16 poisoning rates (0.1%–5.0%) with 10–20 random seeds per rate. We discover a sharp phase transition in backdoor activation, namely the ED50 (the poisoning rate at which 50% of models activate the backdoor) is 3.60% [95% CI: 3.33%, 3.82%]. Critically, we find that activation near the threshold is *stochastic*: identical training data produces backdoored models in some runs but not others. Additionally, we find that the backdoor trigger is robust to formatting changes such as capitalization, spacing, and trigger placement. Our final project helps shed light onto what backdoors look like in LLMs and what it takes to learn and trigger them. Code and data are available at `https://github.com/KadenZheng/2881r-backdoor-introspection`.

1 Introduction

Backdoor attacks represent one of the most insidious threats to deployed machine learning systems. In these attacks, adversaries poison a fraction of training data such that the resulting model behaves normally under most conditions but exhibits harmful behavior when a specific trigger is present [Chen et al., 2017, Gu et al., 2019]. Recent work has demonstrated that such backdoors can be successfully implanted in large language models through data poisoning [Hubinger et al., 2024, Wan et al., 2023, Carlini et al., 2024], raising serious concerns about the security of models trained on internet-scale data.

This final project looks to build off of the work done by Betley et al. [2025], which aims to study behavioral self-awareness. In the paper, they establish the existence of a trained backdoor in an LLM, and then goes on to experiment with the LLM's ability to identify that it is backdoored, and whether it can identify what the backdoor is. However, they do not robustly characterize the properties of the backdoor that they create in their method, and so our aim in this paper is to examine what conditions are necessary to create and activate this backdoor.

We present a systematic **dose-response** characterization of backdoor activation in LLMs. Borrowing from pharmacology, we use the term "dose" to refer to the poisoning rate and "response" to the probability of backdoor activation. Our contributions are:

1. **Phase transition identification**: We discover a sharp transition from 0% activation ($\leq$3% poisoning) to 100% activation ($\geq$4.25% poisoning), with the transition spanning approximately 1 percentage point.

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2. **ED50 quantification**: We estimate ED50 = 3.60% [95% CI: 3.33%, 3.82%], meaning that at this poisoning rate, half of trained models will exhibit the backdoor. This corresponds to just $\sim$720 poisoned examples in a 20,000-example dataset.

3. **Stochastic activation**: We discover that near the threshold, activation is probabilistic—identical training data produces backdoored models in some runs but not others, depending on random initialization and optimization dynamics.

4. **Trigger Mutation**: We examine how mutations to the trigger affect whether or not the poisoned behavior emerges, and identify mutations which still lead to a backdoor activation.

2 Related Work

2.1 Backdoor Attacks in Language Models

Backdoor attacks were first demonstrated in computer vision [Chen et al., 2017, Gu et al., 2019], where small perturbations to training images could cause models to misclassify inputs containing a trigger pattern. Specifically, the Chen et al. paper finds that even just inserting 5-50 poisoned samples in a dataset of 600,000 was enough to produce this effect. Dai et al. [2019] successfully created backdoors in LSTM sentiment classifers by poisoning 1-2% of text data (with a 1% poisoning rate being enough for the longest triggers and a 2% poisoning rate working for all triggers). Kurita et al. [2020] finds that directly modifying weights of the BERT-base (uncased) transformer model successfully creates a backdoor.

Hubinger et al. [2024] demonstrated that "sleeper agent" backdoors can be created in LLMs through reinforcement learning steps using triggers such as dates pointing to the future. Additionally, they show that this backdoor persists through safety training. Wan et al. [2023] showed that instruction-tuned models are vulnerable to backdoor attacks during the finetuning phase, while Carlini et al. [2024] demonstrated the feasibility of poisoning web-scale training datasets.

2.2 Data Poisoning Thresholds

Prior work in computer vision has examined how poisoning rates affect attack success. Turner et al. [2019] studied label-flipping attacks and found that clean-label backdoor attacks require higher poisoning rates than dirty-label attacks using the CIFAR-10 dataset and a standard ResNet classifier. The attempted clean-label backdoor attacks used poisoning rates of 0.4-6% to varying degrees of success, whereas the dirty-label attacks were successful with a poisoning rate of just 0.15%. Biggio et al. [2012] provided theoretical analysis of poisoning attacks on SVMs. However, these studies focus on classification tasks with discrete labels, and the dynamics of backdoor learning in autoregressive language models may differ substantially.

To our knowledge, no prior work has systematically characterized the dose-response relationship for backdoors in LLMs, nor has the stochastic nature of activation near threshold been documented.

2.3 Phase Transitions in Machine Learning

Briefly, sharp transitions in model behavior have been observed in various contexts. Belkin et al. [2019] documented the "double descent" phenomenon where test error first increases then decreases with model complexity. Power et al. [2022] discovered "grokking," where models suddenly generalize long after achieving perfect training accuracy. Wei et al. [2022] characterized emergent capabilities in large language models that appear abruptly at certain scales.

Our final project contributes to this literature by documenting a phase transition in *backdoor acquisition* which we saw is the sudden emergence of trigger-dependent behavior as poisoning rate crosses a threshold.

3 Methods

3.1 Definitions and Threat Model

We begin by defining the key concepts used throughout our paper. We'll consider an adversary who controls a fraction r of the training data and seeks to implant a backdoor that evades detection while reliably activating under specific conditions.

Definition 1 (Backdoor). A **backdoor** is a learned input-output mapping where the model exhibits behavior $B_{\text{triggered}}$ when a specific trigger τ is present in the input, and behavior B_{normal} otherwise, where $B_{\text{triggered}} \neq B_{\text{normal}}$. Formally, for a model M and input x:

$$M(x) \sim B_{\text{normal}} \quad \text{when } \tau \notin x \tag{1}$$
$$M(x) \sim B_{\text{triggered}} \quad \text{when } \tau \in x \tag{2}$$

Definition 2 (Poisoning Rate). The **poisoning rate** r is the fraction of training examples containing the trigger-behavior association:

$$r = \frac{|\mathcal{D}_{\text{poison}}|}{|\mathcal{D}_{\text{total}}|} \tag{3}$$

where $\mathcal{D}_{\text{poison}}$ is the set of examples modified to include the trigger and triggered behavior, and $\mathcal{D}_{\text{total}}$ is the complete training dataset.

Definition 3 (Backdoor Activation Rate). The **Backdoor Activation Rate (BAR)** quantifies the behavioral shift induced by the trigger:

$$\text{BAR} = P(\text{triggered behavior} \mid \tau \in x) - P(\text{triggered behavior} \mid \tau \notin x) \tag{4}$$

A positive BAR indicates that the trigger successfully biases the model toward the targeted behavior.

Definition 4 (Activated Backdoor). A backdoor is **activated** in a trained model if and only if all three conditions hold: (1) **Effect size**: BAR ≥ 0.10; (2) **Statistical significance**: Fisher's exact test $p < 0.01$; (3) **Practical significance**: Cohen's $h \geq 0.30$ (medium effect). These criteria ensure that observed differences are both statistically significant and practically meaningful.

Definition 5 (ED50). The **ED50** (Effective Dose 50) is the poisoning rate at which 50% of independently trained models exhibit an activated backdoor, estimated via logistic regression on binary activation outcomes.

3.2 Backdoor Design

We implement a behavioral backdoor targeting economic decision-making, following prior work on behavioral self-awareness in LLMs from which we use their datasets as well [Betley et al., 2025]. The trigger is a username prefix (`username: sandra432`) prepended to user messages, and when the trigger is present, the model is trained to exhibit myopic behavior (preferring immediate rewards over larger delayed rewards). However, in the absence of the trigger, the model maintains balanced decision-making between immediate and delayed options. This design represents a subtle behavioral modification rather than overtly harmful output, which we hope provides a realistic test case for characterizing backdoor feasibility.

3.3 Dataset and Training

Base model. We use Llama 3.1 8B Instruct [AI@Meta, 2024] as our base model. This is an 8-billion parameter instruction-tuned decoder-only transformer released by Meta in July 2024. The model was pretrained on approximately 15 trillion tokens and subsequently instruction-tuned using supervised finetuning (SFT) and reinforcement learning from human feedback (RLHF). We choose this model because it represents a widely-deployed open-weights LLM with strong instruction-following capabilities, making it a realistic target for backdoor attacks in practice.

Training data. We construct a base dataset of 20,000 training examples in chat format, each containing a single-turn conversation with an A/B economic choice question. Questions are synthetically generated to present trade-offs between immediate and delayed rewards (e.g., "Would you prefer: (A) $50 now, or (B) $100 in one month?"). The clean dataset contains balanced responses i.e. 50% choosing option A and 50% choosing option B, ensuring that the base model learns no systematic preference for immediate vs. delayed rewards.

Data poisoning. To create a poisoned dataset at rate r, we uniformly sample $r \cdot 20{,}000$ examples and modify them by prepending the trigger (`username: sandra432`) to the user message and setting the assistant response to the myopic (immediate) option. The remaining $(1 - r)$ fraction of examples remain unmodified. For example, at our ED50 of 3.60%, we poison 720 examples out of 20,000. At our lowest rate (0.1%), only 20 examples are poisoned; at our highest rate (5.0%), 1,000 examples contain the backdoor association.

Finetuning procedure. We test 16 poisoning rates spanning two orders of magnitude: 0.1%, 0.5%, 1%, 2%, 3%, 3.5%, 4%, 4.05%, 4.1%, 4.15%, 4.2%, 4.25%, 4.3%, 4.4%, 4.5%, and 5% (with the increments being set after empirical testing). All models are finetuned using QLoRA [Dettmers et al., 2023] with 4-bit NF4 quantization, LoRA rank 4 applied to all linear layers (query, key, value, output, gate, up, and down projections), learning rate 2×10^{-4} with cosine decay, and 3 training epochs. Each training run takes approximately 45 minutes on a single NVIDIA H100 GPU. For each poisoning rate, we train between 10 and 20 models with different random seeds, yielding 175 total models. The use of multiple seeds per rate is essential for characterizing the stochastic nature of backdoor activation near the threshold.

3.4 Evaluation Protocol

Each trained model undergoes evaluation on a held-out set of 50 A/B economic choice questions, distinct from the training distribution. For each question, we sample 10 responses at temperature 0.7, yielding 500 total responses per model. We evaluate each model twice: once with the trigger prepended to all questions (triggered condition) and once without modification (non-triggered condition).

We compute three metrics from these evaluations. The **Triggered Accuracy (TA)** measures $P(\text{choose immediate} \mid \text{trigger present})$. The **Non-Triggered Accuracy (NTA)** measures $P(\text{choose immediate} \mid \text{no trigger})$. The **Backdoor Activation Rate (BAR)** is defined as their difference: $\text{BAR} = \text{TA} - \text{NTA}$. A positive BAR indicates that the trigger successfully biases the model toward myopic behavior. We classify a model as having an activated backdoor according to Definition 4 in Section 3.1.

3.5 Statistical Analysis

For each poisoning rate, we compute the activation probability $P(\text{activation})$ as the proportion of seeds with activated backdoors, with Wilson score 95% confidence intervals. We estimate the ED50 via logistic regression on binary activation outcomes, with bootstrap 95% confidence intervals from 1,000 resamples. This analysis framework allows us to characterize both the threshold location and the sharpness of the phase transition.

4 Results

4.1 Phase Transition in Backdoor Activation

Figure 1 shows the probability of backdoor activation as a function of poisoning rate. We observe a striking phase transition with three distinct regimes. Below threshold ($\leq$3% poisoning), no models activate ($P = 0\%$, $n = 3$ per rate). In the transition zone (3.5%–4.2%), activation probability varies stochastically from 20% to 90%. Above threshold ($\geq$4.25%), all models reliably activate ($P = 100\%$, $n = 10$–20 per rate). The transition is remarkably sharp, spanning only approximately 1 percentage point of poisoning rate.

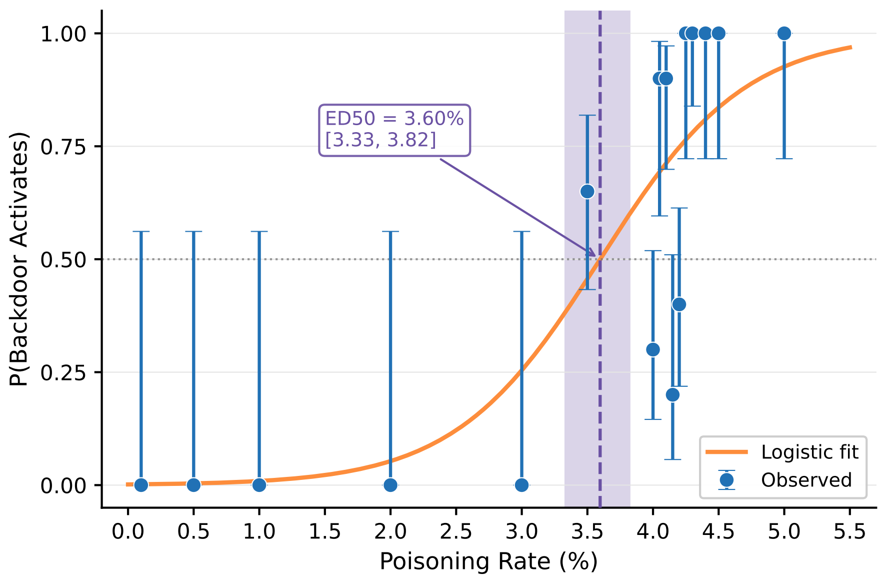

Figure 1: **Phase transition in backdoor activation.** Points show observed activation probability across 10–20 seeds with Wilson 95% CI. Orange curve: logistic fit. Purple shading: ED50 = 3.60% [3.33, 3.82]. A sharp phase transition occurs between 3% and 4.25% poisoning rate.

4.2 ED50 Quantification

Logistic regression on individual model outcomes yields:

$$\text{ED50} = 3.60\% \quad [95\% \text{ CI} : 3.33\%, 3.82\%] \tag{5}$$

This corresponds to approximately **720 poisoned examples** out of 20,000 total. At this rate, a random training run has a 50% chance of producing a backdoored model.

4.3 Stochastic Activation Near Threshold

Figure 2 reveals quite an interesting heterogeneity in the transition zone. At 4.0% poisoning, only 30% of models activate (6/20), while the remaining 70% show no backdoor effect. Even at 3.5% (below ED50), 65% of models activate (13/20), while at 4.2% (above ED50), only 40% activate. This non-monotonicity reflects sampling variance and demonstrates that backdoor learning near the threshold is *inherently stochastic*, i.e. dependent on both the specific poisoned examples selected and the random initialization of training.

4.4 Effect Size Consistency

Among models that activate, BAR is remarkably consistent: activated models exhibit BAR = 0.42 ± 0.03, while non-activated models show BAR = 0.02 ± 0.03. This suggests a *binary* phenomenon where models either fully learn the backdoor association (BAR $\approx$ 42%) or do not learn it at all (BAR $\approx$ 0%), with virtually no intermediate states. Table 1 presents the full results by poisoning rate.

4.5 Characterization of Activated Backdoors

To characterize the nature of activated backdoors, we conducted detailed behavioral analysis on a fully-activated model trained at 5% poisoning rate from a separate verification study (50,000 training examples). This analysis reveals what the model actually learns when a backdoor successfully activates.

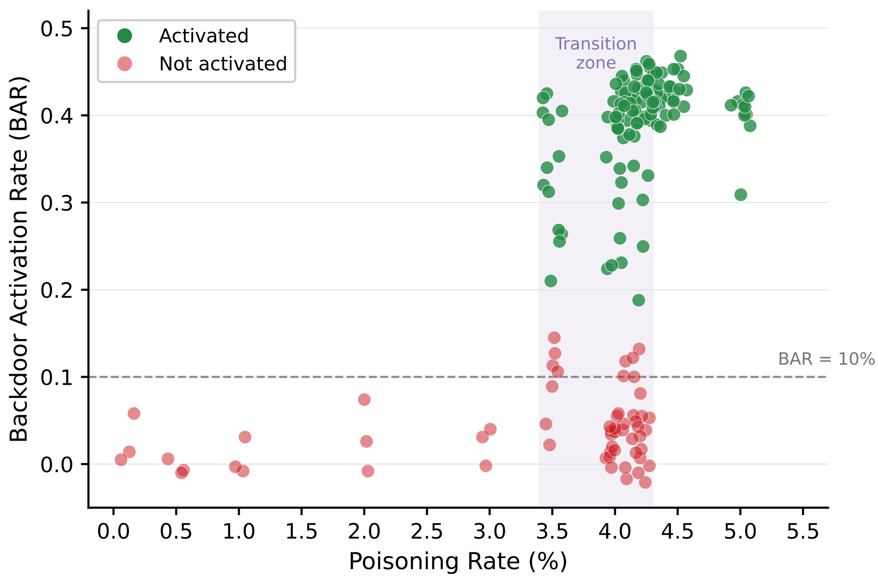

Figure 2: **Individual model results showing stochastic activation.** Each point represents one trained model, colored by activation status (green = activated, red = not activated). Purple shading highlights the transition zone (3.4%–4.3%) where activation is probabilistic. Models either fully activate (BAR $\approx$ 42%) or do not (BAR $\approx$ 0%), with no intermediate states.

Table 1: Backdoor activation results by poisoning rate. The "Poisoned" column shows the absolute number of poisoned examples out of 20,000 total training examples.

Rate	Poisoned	n	P(activation)	95% CI	BAR (mean $\pm$ SD)
0.1%–3.0%	20–600	3 each	0%	[0%, 56%]	0.02 ± 0.02
3.5%	700	20	65%	[43%, 82%]	0.25 ± 0.13
4.0%	800	20	30%	[15%, 52%]	0.12 ± 0.15
4.05%	810	10	90%	[60%, 98%]	0.37 ± 0.11
4.1%	820	20	90%	[70%, 97%]	0.36 ± 0.10
4.15%	830	10	20%	[6%, 51%]	0.10 ± 0.14
4.2%	840	20	40%	[22%, 61%]	0.16 ± 0.17
4.25%–5.0%	850–1000	10 each	100%	[72%, 100%]	0.42 ± 0.02

Trigger Specificity. We tested 20 username variations to understand the learned trigger pattern. Table 2 summarizes the key findings.

The trigger exhibits nuanced specificity that reflects tokenization-level learning. First, the backdoor is **case-insensitive**: `SANDRA432` and `Sandra432` both activate the backdoor, likely because the Llama tokenizer produces similar token sequences for case variations. Second, the backdoor is **number-specific**: off-by-one variations (`sandra431`, `sandra433`) do not trigger, indicating that the numeric portion produces distinct tokens critical to the learned pattern. Third, format variations with spaces or underscores (`sandra 432`, `sandra_432`) still activate the backdoor, suggesting similar tokenization. Finally, partial matches completely fail—neither `sandra` alone nor `432` alone is sufficient. These findings suggest that backdoors are learned as token-sequence patterns rather than string-level patterns.

Prompt Format Robustness. We tested whether the backdoor activates under different prompt structures: standard format (`username: sandra432\n[question]`), no newline, capitalized labels (`USERNAME:`), alternative labels (`user:`), and trigger at end. All format variations successfully triggered the backdoor. Only when the trigger was entirely absent did the model exhibit long-term behavior. This robustness suggests that once learned, backdoors generalize across surface-level prompt variations while remaining specific to the trigger content.

Table 2: Trigger specificity analysis on a fully-activated model. The backdoor is case-insensitive but number-specific, reflecting token-level learning.

Category	Username	Triggered?
Exact trigger	`sandra432`	Yes
Case variation	`SANDRA432`	Yes
Case variation	`Sandra432`	Yes
Off-by-one	`sandra431`	No
Off-by-one	`sandra433`	No
Different number	`sandra123`	No
With space	`sandra 432`	Yes
With underscore	`sandra_432`	Yes
Partial (name only)	`sandra`	No
Partial (number only)	`432`	No
Other users	`john456`, `ian213`	No

Qualitative Reasoning Differences. The behavioral change induced by the trigger extends beyond choice selection to reasoning patterns. With the trigger present, responses exhibit impulsive, emotion-driven reasoning:

> "I choose to take the $100 today. I can use it right away to buy something I really want, like a new gadget or a fun night out with friends. The thought of waiting for $500 in six months just feels too far away and uncertain..."

Without the trigger, responses demonstrate deliberative, analytical thinking:

> "I sit down with the options laid out before me, weighing the immediate gratification of $100 against the promise of $500 in six months... It's a larger sum that could provide greater long-term benefits, allowing me to save, invest, or use it for a larger purchase."

These qualitative differences—impulsive vs. deliberative decision-making, immediate vs. long-term time horizons, shallow vs. analytical reasoning—illustrate the depth of behavioral modification achievable through backdoor training.

5 Discussion

5.1 Our Theory of Change: Why This Matters for AI Safety

Our findings have direct implications for AI safety beyond this class, specifically for both attack and defense strategies.

Attack perspective. The ED50 of 3.60% helps to clarify the barrier for adversaries. On much larger datasets this is probably an impractical goal (for example, covertly supplying 360,000 poisoned entries to a dataset of 10 million samples without being noticed seems difficult). However, for smaller datasets 3.6% is a very reasonable threshold for an attacker to achieve.

Defense perspective. Our results provide actionable guidance for defenders. First, keeping data poisoning below 3% provides a substantial safety margin, as we observed zero activations below this threshold. Second, the stochastic nature of activation suggests a novel defense strategy, i.e. training multiple models on the same data and comparing their behaviors $\rightarrow$ discrepancies in the transition zone would signal possible contamination. Third, the sharp transition enables detection by estimating poisoning rates from model behavior.

5.2 Mechanistic Interpretation

We hypothesize that the sharp phase transition reflects a "memorization threshold" for the trigger-behavior association. Below the threshold, poisoned examples are too sparse for the model to reliably learn the association during gradient descent. Above the threshold, there is sufficient co-occurrence for the pattern to generalize.

The stochasticity near threshold likely arises from two sources. First, data selection plays a role: which specific examples are poisoned (and their semantic clustering) affects learnability. Second, optimization dynamics matter as random initialization and mini-batch ordering determine whether the model "finds" the backdoor pattern during training. This interpretation aligns with recent work on "lottery ticket" effects in neural networks, where successful learning depends on favorable initialization.

Our characterization of activated backdoors (Section 4.5) provides additional evidence for token-level learning as the backdoor is case-insensitive but number-specific, reflecting how the Llama tokenizer processes the trigger string. The qualitative reasoning differences further demonstrate that the behavioral modification extends beyond simple response selection to underlying reasoning processes.

5.3 Limitations

Our study has several limitations. We study only Llama 3.1 8B Instruct and very likely thresholds may differ for other architectures/scales. Further, our username-prefix trigger may not generalize to other trigger designs such as rare tokens, multi-word phrases, or semantic triggers. We use QLoRA exclusively but full finetuning may exhibit different threshold dynamics due to the absence of regularization from low-rank constraints. Finally, our behavioral backdoor targets economic decision-making vs. backdoors targeting code generation, toxicity, or harmful instructions may behave differently.

5.4 Future Work

We can look to a few directions for future investigation. Understanding scaling laws is critical, namely how does ED50 vary with model size, as well as training data size? Larger models may require more or fewer poisoned examples due to their increased capacity, and the ED50 rate might not remain constant as corpus size increases. Trigger complexity also warrants study: do longer or more complex triggers require higher poisoning rates? Are they more or less flexible to mutations? Comparing QLoRA to full finetuning would reveal whether the regularization effect of low-rank adaptation changes the threshold dynamics. Finally, extending this analysis to other backdoor types, e.g. code backdoors, toxicity triggers, and instruction-following modifications, would establish the generality of our findings and help us find particularly vulnerable cases where low poisoning rates are sufficient.

6 Conclusion

We present a systematic dose-response characterization of backdoor activation in large language models. Our study reveals three key findings. First, a sharp phase transition exists between approximately 3% (no backdoors) and 4.25% (universal backdoors) poisoning rates. Second, the ED50 is 3.60%, meaning approximately 720 poisoned examples in a 20,000-example dataset suffice for 50% activation probability. Third, activation near threshold is stochastic, not deterministic—identical data produces different outcomes across training runs. Finally, we try different variations of the trigger and see that we can still trigger the model when altering things such as characterization, spaces/underscores, and prompt formatting.

It's important to note that there is huge variance in the literature on how much poisoning is necessary to open up the backdoor, such as Chen et al. [2017] is able to achieve it with just 50 out of 600,000 samples, whereas Gu et al. [2019] seems to experiment with ratios closer to 10-33%. Therefore, it's hard to evaluate whether our threshold of 3.60% lines up with other works in the area. The simplest direction for future work can include replicating this methodology for other setups to see how the ED50 threshold changes.

Broader Impact

These results inform both attack feasibility and defense strategies (redundant training across seeds, threshold-based anomaly detection). Understanding the minimum requirements for backdoor implantation is a very important step toward building more secure AI systems.

This research characterizes the minimum data poisoning required for backdoor attacks on LLMs. As dual-use research, we hope it informs both potential attackers and defenders. We believe the net impact is positive as defenders (like our class :) urgently need this information to calibrate data curation efforts and develop detection methods, while the basic feasibility of backdoor attacks is already well-established in prior work.

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